

Decode the Chart

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For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

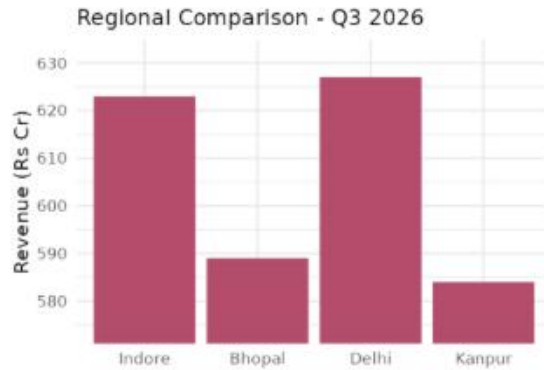


Chart 1

Budget Allocation by Department (14 categories)



Chart 2

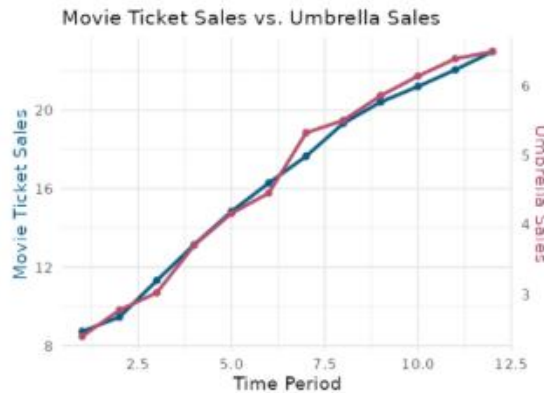


Chart 3

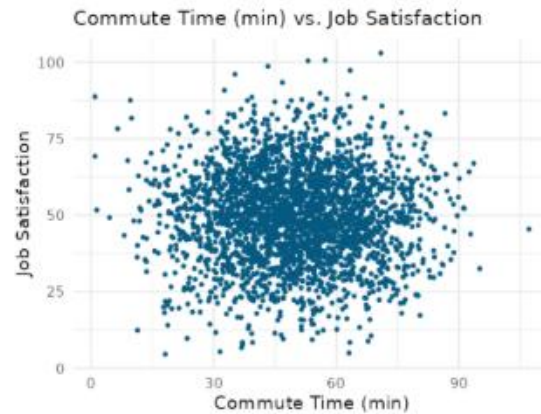


Chart 4

Your notes:

chart-1 Regional comparison

* Delhi (about 627cr) and indore (about 623cr) look vastly bigger than Bhopal (about 589cr) and kanpur (about 584cr). On the picture, Delhi's bar looks roughly 5 to 6 times taller than kanpur's.

The Y-axis starts at about 578, not 0. cutting off the bottom of the bars exaggerates small differences. In reality Delhi is only about 7% higher than kanpur (627 vs 584), and indore is only about 6% higher than Bhopal. Bar length should be proportional to the value, so bars must be zero.

chart-2 Budget Allocation by Department

With 14 slices, many are thin and many colours look like (several blues, greens and pink). There are no labels or percentages, so we cannot tell which slice is 3rd largest, or compare slices that are not next to each other. Humans judge angle and area poorly, and the legend-matching efforts is too high.

chart-3 movie Ticket sales vs umbrella sales

The two lines are use separate Y-axis's that were scaled so the lines sit on top of each other. The overlap comes from axis scaling, not from a real relationship. Both quantities simply trend upward over time, and trending together is not causation; buying a movie ticket does not cause umbrella sales. unrelated things that both grow over time will always look correlated.

Chart - 4 - Commute Time vs Job Satisfaction

There are about 2000 or more fully opaque dots. In the middle 3 to 7 min, satisfaction about 30 to 70, the points pile on top of each other into a solid blob, so, we cannot see where the data is really concentrated or how dense it is.